



CompTIA Security+ (SY0-701) 5-Week Syllabus

This syllabus outlines a detailed 5-week study roadmap for the CompTIA Security+ (SY0-701) exam. Each week focuses on one of the five primary domains, evenly distributed across two instructional days: a Wednesday PM session and a Saturday AM session, including concepts for each objective and a weekly hands-on milestone.

Week 1: General Security Concepts (Domain 1.0)

Wednesday PM Session

1.1 Compare and contrast various types of security controls.

- Categories
- Control types

1.2 Summarize fundamental security concepts.

- CIA Triad
- Non-repudiation
- AAA
- Gap analysis
- Zero Trust
- Physical security
- Deception/disruption

Saturday AM Session

1.3 Explain the importance of change management processes and the impact to security.

- Business processes
- Technical implications
- Documentation

1.4 Explain the importance of using appropriate cryptographic solutions.

- PKI
- Encryption
- Tools
- Obfuscation
- Hashing
- Salting
- Digital signatures
- Key stretching
- Blockchain



Week 1

- Certificates

Weekly Hands-on Milestone

- Design a basic secure system architecture diagram
- Show data at rest, in transit, and in use with correct protections labeled
- Mark where encryption, hashing, and PKI components are applied



Week 2: Threats, Vulnerabilities, and Mitigations (Domain 2.0)

Wednesday PM Session

2.1 Compare and contrast common threat actors and motivations.

- Threat actors
- Attributes of actors
- Motivations

2.2 Explain common threat vectors and attack surfaces.

- Message-based
- Image/File-based
- Voice call
- Removable device
- Vulnerable software
- Unsupported systems
- Unsecure networks
- Open service ports
- Default credentials
- Supply chain
- Social engineering

2.3 Explain various types of vulnerabilities.

- Application
- OS-based
- Web-based
- Hardware
- Virtualization
- Cloud-specific
- Supply chain
- Cryptographic
- Misconfiguration
- Mobile device
- Zero-day

Saturday AM Session

2.4 Given a scenario, analyze indicators of malicious activity.

- Malware attacks
- Physical attacks
- Network attacks
- Application attacks
- Cryptographic attacks
- Password attacks
- Indicators

2.5 Explain the purpose of mitigation techniques used to secure the enterprise.

- Segmentation
- Access control
- Application allow list
- Isolation
- Patching
- Encryption



Week 2

- Monitoring
- Least privilege
- Configuration enforcement
- Decommissioning
- Hardening techniques

Weekly Hands-on Milestone

- Conduct a threat modeling exercise on a simple web application
- Map and identify all attack surfaces and exposure points
- Propose and document mitigations for each identified exposure



Week 3: Security Architecture (Domain 3.0)

Wednesday PM Session

3.1 Compare and contrast security implications of different architecture models.

- Architecture concepts
- Network infrastructure
- Centralized/decentralized
- Containerization
- Virtualization
- IoT
- ICS/SCADA
- RTOS
- Embedded systems
- High availability

3.2 Given a scenario, apply security principles to secure enterprise infrastructure.

- Infrastructure considerations
- Device placement
- Network appliances
- Secure access
- Selection of controls

Saturday AM Session

3.3 Compare and contrast concepts and strategies to protect data.

- Data types
- Data classifications
- General data considerations
- Methods to secure data

3.4 Explain the importance of resilience and recovery in security architecture.

- High availability
- Site considerations
- Continuity of operations
- Capacity planning
- Testing
- Backups
- Power

Weekly Hands-on Milestone

- Design an enterprise network architecture diagram from scratch
- Include DMZ, internal network, and secure admin access zones
- Correctly place WAF, NGFW firewall layers, and jump servers



Week 4: Security Operations (Domain 4.0)

Wednesday PM Session

4.1 Given a scenario, apply common security techniques to computing resources.

- Secure baselines
- Hardening targets
- Installation considerations
- Mobile solutions
- Wireless security
- Application security
- Sandboxing
- Monitoring

4.2 Explain the security implications of proper hardware, software, and data asset management.

- Acquisition process
- Accounting/Ownership
- Classification
- Asset tracking
- Inventory
- Disposal
- Sanitization
- Data retention

4.3 Explain various activities associated with vulnerability management.

- Identification methods
- Analysis
- Vulnerability response
- Validation of remediation

4.4 Explain security alerting and monitoring concepts and tools.

- Monitoring computing resources
- Activities
- Alert response
- Tools

4.5 Given a scenario, modify enterprise capabilities to enhance security.

- Firewall
- IDS/IPS
- Web filter
- OS security
- Secure protocols
- DNS filtering
- Email security
- FIM
- DLP



Week 4

- NAC
- EDR/XDR

Saturday AM Session

4.6 Given a scenario, implement and maintain identity and access management.

- Provisioning user accounts
- Permission assignments
- Identity proofing
- Federation
- Access controls
- MFA
- Password concepts
- PAM tools

4.7 Explain the importance of automation and orchestration related to secure operations.

- Use cases
- Benefits
- Other considerations

4.8 Explain appropriate incident response activities.

- Process
- Training
- Testing
- Root cause analysis
- Threat hunting
- Digital forensics

4.9 Given a scenario, use data sources to support an investigation.

- Log data
- Data sources

Weekly Hands-on Milestone

- Analyze a sample CVSS report - determine severity score and attack vector
- Identify the exploit path and affected components from the report
- Prioritize patch deployment order based on business risk and exploitability



Week 5: Security Program Management and Oversight (Domain 5.0)

Wednesday PM Session

5.1 Summarize elements of effective security governance.

- Guidelines/Policies
- Standards
- Procedures
- External considerations
- Monitoring
- Governance structures
- Roles and responsibilities

5.2 Explain elements of the risk management process.

- Risk identification
- Risk assessment
- Risk analysis
- Risk register
- Risk tolerance
- Risk strategies
- Risk reporting
- BIA

5.3 Explain the processes associated with third-party risk assessment and management.

- Vendor assessment
- Vendor selection
- Agreement types
- Vendor monitoring

Saturday AM Session

5.4 Summarize elements of effective security compliance.

- Compliance reporting
- Consequences of non-compliance
- Compliance monitoring
- Privacy

5.5 Explain types and purposes of audits and assessments.

- Attestation
- Internal
- External
- Penetration testing



Week 5

5.6 Given a scenario, implement security awareness practices.

- Phishing
- Anomalous behavior
- User guidance/training
- Reporting
- Execution

Weekly Hands-on Milestone

- Calculate SLE and ALE from a given asset value, exposure factor, and ARO
- Classify a given risk scenario as qualitative vs quantitative and justify your choice
- Draft a sample vendor risk assessment checklist using SLA, NDA, and MOU criteria